

## Practical 4

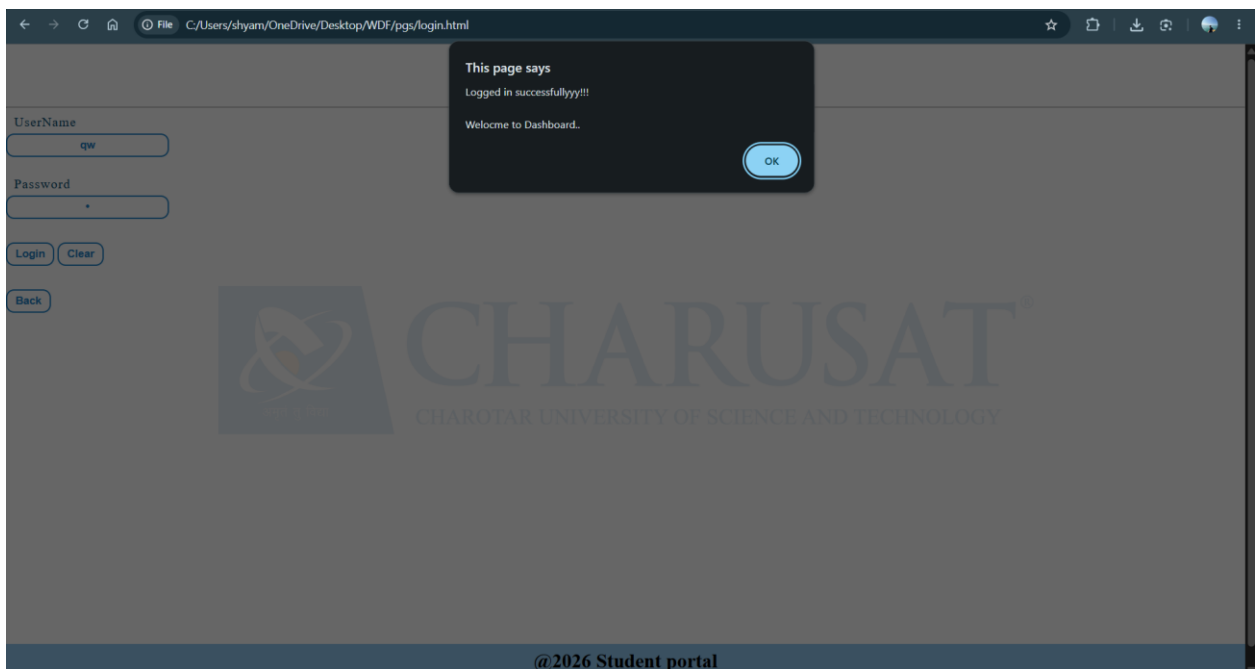
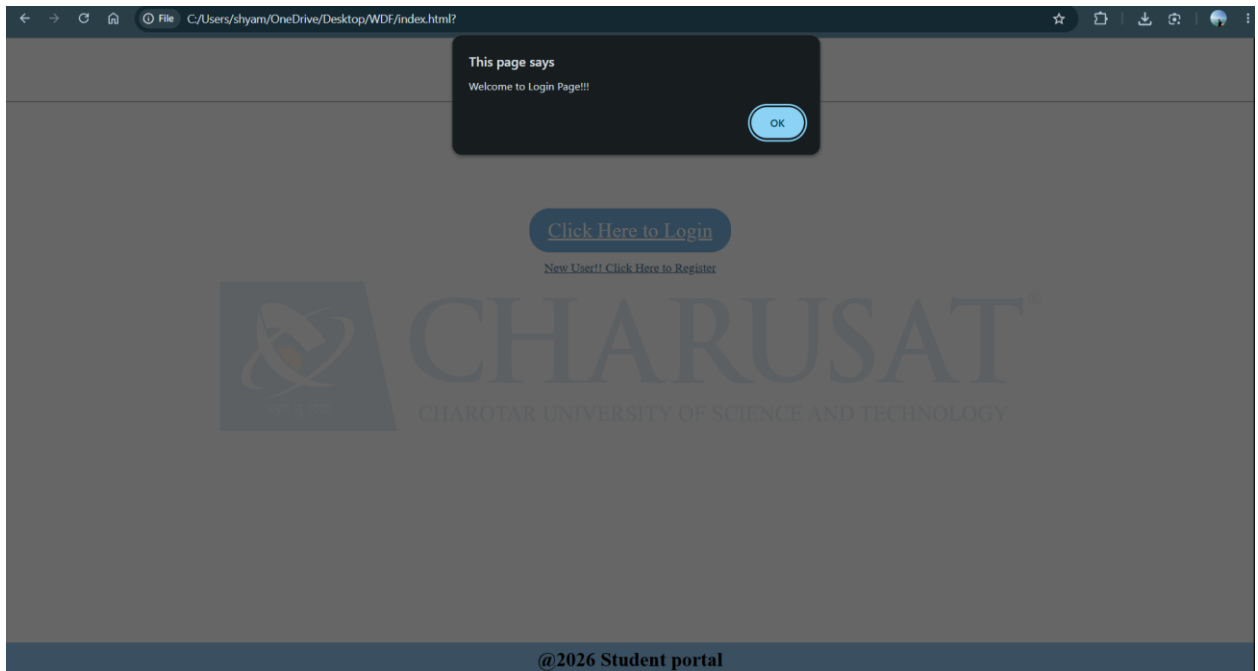
Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

### A. Theory :

Practical 4 was about making the StudentHub Portal interactive using JavaScript, DOM manipulation, event handling and UI components. The purpose was to take the HTML and CSS pages created in the previous practicals and add functionality to them using JavaScript. The practical required working with webpage elements through the DOM, responding to user actions such as button clicks and form submission, changing CSS classes, and creating interactive features such as a hamburger menu, dark mode, FAQ section, modal popup, notification banner and content slider. JavaScript was also used to handle the login form and provide different dashboard access according to the selected user type. The practical helped in understanding how JavaScript connects HTML and CSS to create a dynamic and interactive website.

### B. Implementation/ Output/Result :

In this practical, I added JavaScript functionality to my StudentHub Portal. I implemented a login system where the user selects Student, Faculty or Admin and enters the corresponding username and password. According to the selected role and valid credentials, the user is redirected to the appropriate dashboard. I also implemented a registration check for password requirements and password confirmation. For the user interface, I added a hamburger menu that can show or hide the dashboard navigation, a Dark Mode and Light Mode feature, and used local storage so that the selected theme remains saved after refreshing the page. I also implemented a collapsible FAQ section, a modal popup for additional information, a notification banner that can be closed, and a content slider with Previous and Next buttons. DOM methods such as `getElementById()`, `querySelector()`, and `querySelector()` were used to access webpage elements, while `addEventListener()` was used to respond to user actions.

**C. Evidence of Work (Screenshots):**

← → ↻ 🏠 File C:/Users/shyam/OneDrive/Desktop/WDF/pgs/register.html ☆ 📄 ⬇️ 🔄 🌐

This page says  
Registration Done successfully!!!  
OK

UserName

Set Password

Re-enter Password



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## Student Hub Portal



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**D. GitHub Repository Link:**

<https://github.com/ShyamPatel9/Student-Hub-Portal>

**E. Challenges Faced:**

While implementing Practical 4, one of the main challenges was understanding how JavaScript connects with HTML elements through the DOM. Initially, some functions did not work because the JavaScript had to find the correct element IDs and class names from the HTML. Another challenge was handling the navigation paths between the Student, Faculty and Admin dashboard folders because relative paths depend on the location of each HTML file. Implementing the hamburger menu also required understanding how to add and remove a CSS class using `classList.toggle()`. Dark mode required understanding both JavaScript class manipulation and `localStorage` so that the selected theme could be remembered. Another challenge was making the same JavaScript file work on different pages without causing errors when a particular element did not exist on that page.

**F. Learning Outcomes:**

After completing Practical 4, I understood how JavaScript can be used to make a website interactive. I learned how to select HTML elements using the DOM and respond to user actions using event listeners. I learned how CSS classes can be controlled using JavaScript to show, hide or change webpage elements. I also understood how features such as dark mode, menus, FAQ sections, popups and sliders can be created using JavaScript. I learned the basic use of `localStorage` for saving user preferences and understood how external JavaScript can be reused across multiple webpages. Overall, this practical helped me understand how HTML, CSS and JavaScript work together in a real website.

**Faculty Remarks:**